Experiment ‘p_5_9_c_cylinder_volume’ Results

June 4, 2026

Experiment outcome: SUCCESS
Bad responses: 0
Responses containing `assume` **:** 0
Responses containing `expect` **:** 0
Resolution attempts: 1
Hard fails (resolution): 0
Soft fails (resolution): 0
Verification attempts: 1

Problem Specification

Problem name: p_5_9_c_cylinder_volume
Natural language statement: Write a method that computes the volume of a cylinder with a circular base, given radius and height.
Method signature: p_5_9_c_cylinder_volume(r: real, h: real) returns (volume: real)

Ensures

- `volume == 3.14159 * r * r * h`

GenAI interactions

Below you will find all interactions between the 'user' (program) and the 'assistant' (OpenAI).

Program → GenAI

```
You are given the following task to perform in Dafny:

Write a method that computes the volume of a cylinder with a circular base, given radius
    ↪ and height.

The signature should be:

p_5_9_c_cylinder_volume(r: real, h: real) returns (volume: real)

The method should respect the following contract:

ensures volume == 3.14159 * r * r * h

Produce and show only the Dafny body of this method, including the curly braces that
    ↪ surround it. Do not show the signature nor contract. You must not use 'assume' nor
    ↪  'expect' anywhere in your code.
```

GenAI → Program

System fingerprint: null
ID: chatcmpl-DmzIeYCZmMc1YVfLl0WmmBDQWVmzr
Model: gpt-5.5-2026-04-23
Created at: 1780567752
Finish reason: stop
Usage: 294 tokens in, and 117 tokens out

```
{
  volume := 3.14159 * r * r * h;
}
```

Final Program

```
method p_5_9_c_cylinder_volume(r: real, h: real) returns (volume: real)
       ensures volume == 3.14159 * r * r * h
{
  volume := 3.14159 * r * r * h;
}
```

Total Token Usage

Input tokens: 294
Output tokens: 117
Reasoning tokens: 81
Sum of 'total tokens': 411

Experiment Timings

Overall Experiment started at 1780567752187, ended at 1780567785926, lasting 33739ms (33.74 seconds)
Iteration #1 started at 1780567752187, ended at 1780567785926, lasting 33739ms (33.74 seconds)